

上 海 交 通 大 学 试 卷 (A 卷)

(2021 至 2022 学 年 第 2 学 期 期 末 考 试)

班级号 _____ 学号 _____

姓名 _____

课程名称 _____ 泛函分析 _____

成绩 _____

1. (10 points) Suppose that $\mathbf{M}$ is a mapping from a Hilbert space H onto itself and preserves the distance of any pair of points, i.e.,

$$\|\mathbf{M}(x) - \mathbf{M}(y)\| = \|x - y\| \quad \text{for all } x, y \in H.$$

If, in addition, $\mathbf{M}(0) = 0$, then prove that $\mathbf{M}$ is a linear map.

我承诺，我将
严格遵守考试纪律。

承诺人: _____

题号	一	二	三	四	五	六	七	八
得分								
批阅人(流水阅)								

2. (16 points) On the space $L^2([0, 1])$, consider the integral operator $(\Lambda u)(t) = \int_0^t u(s)ds$.

(a) Prove that the operator Λ is compact.

(b) Compute the adjoint operator Λ^* .

(c) Compute the spectrum of Λ .

3. (14 points) Please answer the following problems with true or false.

- (a) Is it true that every Hilbert space is separable?
- (b) Is it true that every bounded linear operator on a Banach space has an eigenvalue?
- (c) Let X and Y be two Banach spaces and Λ be a compact map from X to Y . Assume that $l_n \rightharpoonup l$ in Y^* . Is it true that $\Lambda' l_n \rightarrow \Lambda' l$ in X^* ?
- (d) Is it true that every linear space with a metric is a normed linear space?
- (e) Let S be a subset of the Banach space X . Is it true that $\text{Span}\{S\}$ is always a closed set?
- (f) Let Λ be a bounded linear operator from a Banach space X into itself. Then the set for all the spectrum of Λ must be bounded. Is it true?
- (g) Every finite dimensional normed linear space over the reals is complete.

4. (10 points) Let H be a Hilbert space over $\mathbb{C}$ and A be a bounded linear operator from H into itself. Suppose that A satisfies

$$(Ax, x) \geq 0 \quad \text{for all } x \in H.$$

- (a) Prove that A is a symmetric operator.
- (b) If A is also compact, prove that all eigenvalues of A are nonnegative.

5. (12 points) Let X be a Banach space.

- (a) Let $S \subset X$ be a closed convex set and assume $y \notin S$. Prove that there exists a bounded linear functional $\varphi \in X^*$ such that

$$\varphi(y) < \inf_{x \in S} \varphi(x).$$

- (b) Consider a weakly convergent sequence $\{x_n\} \subset X$, i.e., $x_n \rightharpoonup y$. Let $S = \overline{\{x_n; n \geq 1\}}$ be the smallest closed convex set containing all points x_n . Prove that $y \in S$.

6. (14 points) Please answer the following problems with true or false.

- (a) Let $\mathbb{N} = \{1, 2, 3, \dots\}$ with the metric $d(x, y) = |x - y|$. Then $\mathbb{N}$ is a set of the second category. Is it true?
- (b) Let X be a Banach space and l is a nonzero bounded linear functional on X . Is it true that l must be an open mapping?
- (c) Let $H = L^2(\mathbb{R})$ and $S = \{f \mid \int_{\mathbb{R}} |f|^2 dx \leq 1\}$. Is it true that S is not a compact set in H ?
- (d) Is it true that a bounded sequence in $L^\infty(\mathbb{R})$ must have a weak-* convergent subsequence?
- (e) Let S be a subset of the Hilbert space H . Then $S^\perp$ must be a closed set. Is it true?
- (f) A Hilbert space H must be reflexive. Is it true?
- (g) Let $\{f_n\}$ satisfy $\int_{\mathbb{R}} |f_n(x)| dx = 1$ for all $n \in \mathbb{N}$. Then $\{f_n\}$ has a weakly convergent subsequence in L^1 . Is it true?

7. (10 points) Given a nonempty closed, convex subset K of a Hilbert space H , and a point x in H , prove that there is a unique point y in K that is closer to x than any other point of K .

8. (14 points) Let X and Y be two Banach spaces. Suppose that $\mathbf{T}$ and $\{\mathbf{T}_n\}_{n=1}^{\infty}$ are bounded invertible linear mappings from X to Y . Suppose that

$$\|\mathbf{T}_n x - \mathbf{T}x\| \rightarrow 0 \quad \text{for all } x \in X.$$

Prove that T is invertible and

$$\|\mathbf{T}_n^{-1}y - \mathbf{T}^{-1}y\| \rightarrow 0 \quad \text{for all } y \in Y$$

if and only if there exists a constant C such that

$$\|\mathbf{T}_n^{-1}\| \leq C \quad \text{for all } n \in \mathbb{N}.$$

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